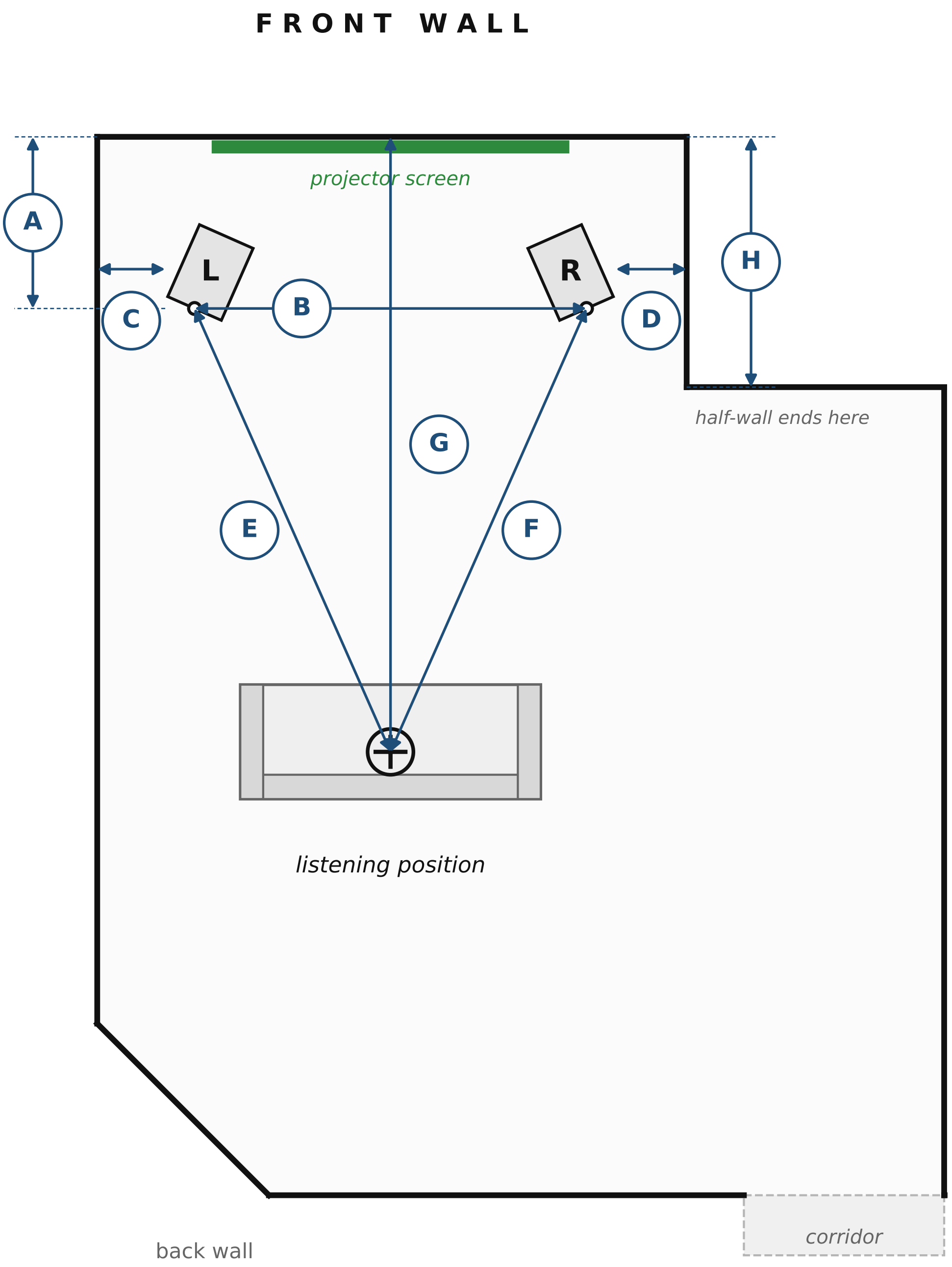
Listening room — measurement sheet

date ______________

B&W Nautilus 801 · 120.blue profile · plan view, not to scale in detail — measure everything below before sweeping



A	tweeter → front wall	______________	cm	*nominal 120 cm for this profile*
B	tweeter → tweeter (speaker separation)	______________	cm	
C	left tweeter → left side wall	______________	cm	*to the TWEETER, not the cabinet*
D	right tweeter → right side wall	______________	cm	*the half-wall, not the far wall*
E	listening position → left tweeter	______________	cm	
F	listening position → right tweeter	______________	cm	
G	listening position → front wall	______________	cm	
H	front wall → end of the half-wall	______________	cm	*decides the right reflection*

mic height _________ mic mounting _________ mic → sofa back _________ ear height _________

toe-in _________ speaker angle _________ anything within 1 m of a speaker _________

Mic mounting is a first-order variable: a mic on a cushion instead of a stand moved the floor bounce from 2.0 to ~1.1 ms and added up to 5.6 dB below 2 ms — larger than any panel change (measured 2026-08-10).

Notes

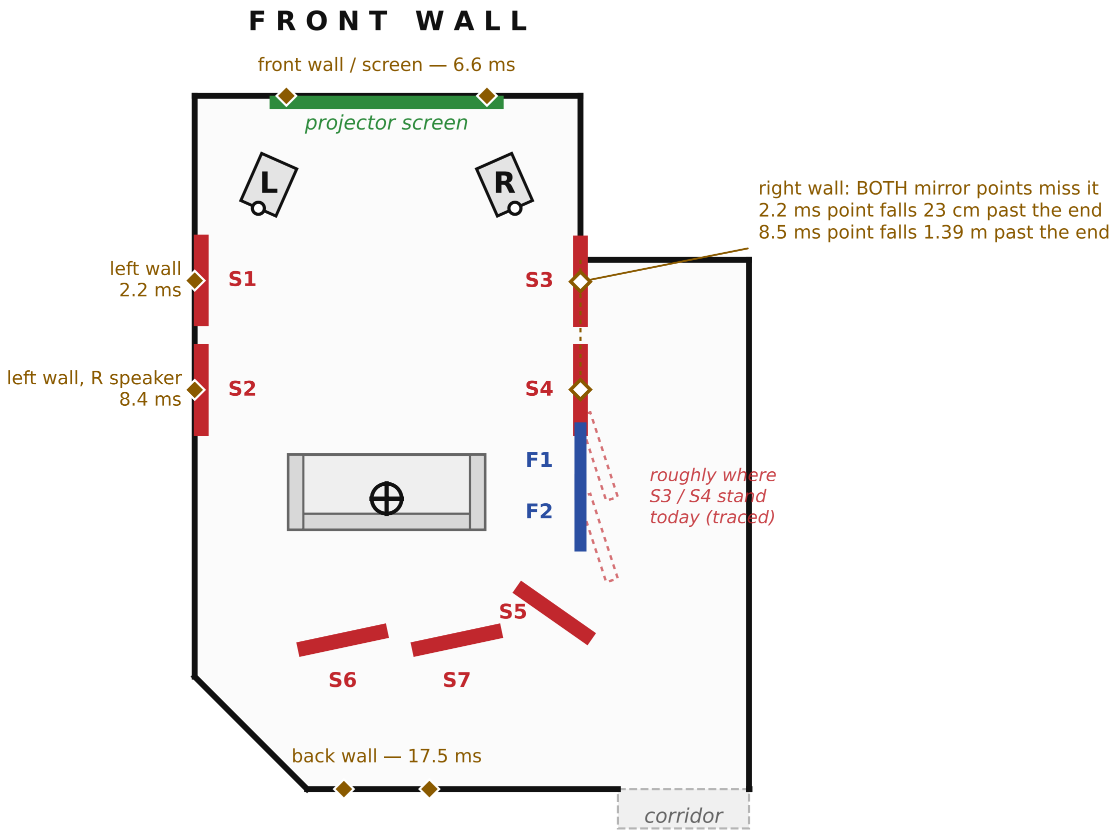
Measured 2026-08-10: tweeters 120 cm from the front wall, 2.74 m apart, 0.68 / 0.70 m to the side walls (to the TWEETER); listener 4.30 m; half-wall 1.75 m. Those sum to 4.120 m against the 4.186 m decoded from room.png. Length 7.40 m and the 1.4 m corridor opening are still from the sketch.

Panel positions & reflection points

date ______________

Targets are COMPUTED mirror points — the panels are drawn there, not where they stand today.

■ GIK freestanding screen ■ pyramidal foam ■ specular reflection point ■ projector screen



panel	TARGET — computed	the reflection it treats	as found	action
S1	left wall, y = 1.97 m	L speaker off the left wall, 2.2 ms	_________ cm	*already covers it (edge 1.77)*
S2	left wall, y = 3.14 m	R speaker off the left wall, 8.4 ms	_________ cm	*on the wall, ±24 cm*
S3	half-wall line, y = 1.98 m	R speaker mirror point, 2.2 ms	_________ cm	*bring it here*
S4	half-wall line, y = 3.14 m	L speaker mirror point, 8.5 ms	_________ cm	*bring it here*
F1	half-wall line, y = 3.90 m	escape into the open side	_________ cm	*behind the screens*
F2	half-wall line, y = 4.45 m	escape into the open side	_________ cm	*behind the screens*
S5	right rear, y ≈ 5.5 m	corridor mouth	_________ cm	*leave*
S6	behind seat, x ≈ 1.6 m	back wall from L, 17.5 ms	_________ cm	*leave*
S7	behind seat, x ≈ 2.8 m	back wall from R, 17.5 ms	_________ cm	*leave*

Configuration measured

- ☐ baseline — as it was: foam at y 2.0 / 2.9 m, S3 / S4 out in the open (measure this first)
- ☐ as drawn — continuous pseudo-wall, 1.75 → 4.6 m
- ☐ as drawn, plus screens in front of the projector screen (1.05 m off centre each side)
- ☐ other ________________________________

Result — early energy relative to direct, 300 Hz – 8 kHz (NOTES §8)

window	L	R	L − R	was, 120 cm (L / R)
0.3 – 2 ms	________	________	________	*−7.9 / −7.3*
2 – 5 ms	________	________	________	*−4.6 / −5.3*
5 – 10 ms	________	________	________	*−9.9 / −8.1*
10 – 20 ms	________	________	________	*−8.5 / −3.2*
EDT 250 Hz	________	________	________	
EDT 500 Hz	________	________	________	

y is the point on the WALL where the ray reflects. Panels lie ON the wall (NOTES §9: flat on the wall at the mirror point is correct for a specular travelling wave). The ray then crosses a 10 cm panel over a 23 cm window at the 1.97 m target, so a 60 cm panel is right if centred within ±19 cm. Fresnel is the bigger constraint: full absorption needs 1.46 m of panel at 500 Hz. Do not judge panels on the bass; change one thing per measurement.